A subgraph density obstruction to biplanarity, and the thickness of inflated cycles

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Abstract

A graph is biplanar if its edges can be covered by two planar subgraphs. We record an elementary obstruction: since every subgraph of a biplanar graph is biplanar, the classical girth-four planar bound applies to *subgraphs*, so a biplanar graph on N vertices can contain no triangle-free subgraph with more than $4N-8$ edges. Unlike density arguments that constrain the whole graph, this one tolerates a host graph dense in triangles. Applied to Catlin's graphs $C_n[K_r]$, the r-inflations of cycles, it gives $\Theta(C_n[K_r]) \geq 3$ for all $r \geq 4$ and $n \geq 4$. For $r \geq 5$ a plain edge count already gives this; the case $r = 4$ does not follow from any edge count, and combined with the arboricity upper bound of Albertson, Boutin and Gethner it yields $\Theta(C_n[K_4]) = 3$ exactly, answering Open Problem 3 of that paper in the case $r = 4$. In particular $C_7[K_4]$, a 10-chromatic graph that survives the Euler bound for biplanarity with two edges to spare, is not biplanar.

1 Background

The Earth–Moon problem, posed by Ringel in 1959, asks for the maximum chromatic number of a biplanar graph. The answer is known to be 9, 10, 11 or 12: Sulanke 9-chromatic biplanar graph gives the lower bound and an averaging argument gives the upper. Closing the gap from below requires a biplanar graph of chromatic number at least 10.

Catlin's graphs are natural candidates. Write $C_n[K_r]$, the *r-inflation* of the cycle C_n, for the lexicographic product: rn vertices grouped into *n fibres* of size r, each fibre a copy of K_r, and all r^2 edges present between fibres adjacent in C_n. Call an edge a *fibre edge* if its ends lie in one fibre and a *join edge* otherwise, so that

$$|V| = rn, \qquad |E_{\text{fibre}}| = n\binom{r}{2}, \qquad |E_{\text{join}}| = r^2 n.$$

Since $C_n[K_r]$ has independence number r when n is odd, its chromatic number is at least $rn/r = n$; for $n = 7$, $r = 4$ it is exactly 10.

The graph $C_7[K_4]$ is therefore the smallest inflation of a cycle that could settle the Earth–Moon problem from below, and it is not eliminated by the standard count: it has 28 vertices and 154 edges against the biplanar ceiling $6 \cdot 28 - 12 = 156$, a margin of two edges. Albertson, Boutin and Gethner [1] ask for the thickness of $C_{2k+1}[r]$ for $k \geq 4$ and $r \geq 4$ as their Open Problem 3; the same question at $k = 3$ is the case relevant to the Earth–Moon bound.

2 The obstruction

We use only the girth-four consequence of Euler's formula: a simple planar graph on $N \geq 3$ vertices with no triangle has at most $2N-4$ edges. Beineke's inequalities for thickness-two graphs [3] are of the same kind, applied to the whole graph.

Lemma 1 (Subgraph density)**.** *Let G be biplanar on N vertices and let $H \subseteq G$ be a triangle-free subgraph. Then $|E(H)| \leq 4N-8$.*

Proof. Fix planar subgraphs G_1, G_2 covering $E(G)$. Each $H_i = H \cap G_i$ is planar, being a subgraph of a planar graph, and triangle-free, being a subgraph of H. Hence $|E(H_i)| \leq 2N-4$, and $|E(H)| \leq |E(H_1)| + |E(H_2)| \leq 4N-8$. □

The point of stating it for a subgraph is that G itself may be full of triangles. Existing density obstructions to biplanarity constrain the host graph's own girth, and so say nothing about graphs like $C_n[K_r]$, in which every fibre is a clique.

The device is not particular to this problem. Applying a global bound to a restriction of an object, chosen so that the interfering structure cancels, is the method behind the subset rank obstruction of [2]: there one restricts a module to an arbitrary subset S of the unknown vertices and sums over it, whereupon the generators internal to S cancel in pairs and a rank condition invisible in the whole object appears. Here the fibre edges play the role of the internal generators — they carry every triangle, and restricting to the join edges is what removes them. What follows is that one application.

Theorem 1. *For all $n \geq 4$ and $r \geq 4$, $\Theta(C_n[K_r]) \geq 3$.*

Proof. Let J be the spanning subgraph of $C_n[K_r]$ consisting of the join edges. A triangle in J would have its three vertices in three distinct fibres, pairwise adjacent in C_n; since C_n has no triangle for $n \geq 4$, J is triangle-free. If $C_n[K_r]$ were biplanar then Lemma 1 with $N = rn$ would give

$$r^2 n \;=\; |E(J)| \;\leq\; 4rn - 8,$$

that is $rn(r-4) \leq -8$, which fails for every $r \geq 4$ and $n \geq 1$. For $r = 4$ the two sides are $16n$ and $16n-8$, so the inequality fails by exactly 8 independently of n. □

Remark. For $r \geq 5$ the conclusion also follows from the plain edge count: $C_n[K_r]$ has $n\big(\binom{r}{2} + r^2\big)$ edges against a biplanar ceiling of $6rn - 12$, and these cross at $r = 5$. The case $r = 4$ is the one no edge count reaches — $22n$ edges against a ceiling of $24n - 12$ — and it is the case relevant to the Earth–Moon problem.

r	join edges	capacity $4rn-8$	all edges	vs. ceiling $6rn-12$
2	$4n$	$8n-8$	$5n$	under
3	$9n$	$12n-8$	$12n$	under
4	$16n$	$16n-8$	$22n$	under (ceiling $24n-12$)
5	$25n$	$20n-8$	$35n$	over
6	$36n$	$24n-8$	$51n$	over

Table 1: Join edges against the capacity of Lemma 1. The bound bites from $r = 4$; the total edge count only bites from $r = 5$.

Corollary 1. $\Theta(C_n[K_4]) = 3$ *for all* $n \geq 4$. *In particular* $C_7[K_4]$ *is not biplanar.*

Proof. Theorem 1 gives the lower bound. For the upper bound, $C_n[K_4]$ is the closed 2-blowup of $C_n[K_2]$, which is 5-regular on $2n$ vertices and so has arboricity 3 by Nash-Williams; Albertson, Boutin and Gethner [1] show that the closed 2-blowup of a graph of arboricity a has thickness at most a. □

Since $C_5[K_4]$ already fails the plain edge count, the first case of the corollary with any content is $n = 6$, and the first odd case is $n = 7$.

3 Verification

The following were checked by computer; scripts are available from the author.

- In $C_7[K_4]$, of all 3276 vertex triples none spans a triangle of join edges alone, confirming the triangle-freeness of J directly rather than through C_n.
- The counts are as stated: 28 vertices, 42 fibre edges, 112 join edges, 154 edges in all, against a per-layer join capacity of $2 \cdot 28 - 4 = 52$ and a two-layer capacity of 104.
- The argument reproduces the non-biplanarity of $C_5[K_4]$ (80 join edges against a capacity of 72), which was previously known.
- Four biplanar decompositions found by SAT for path sub-cases — six fibres in a path, and seven fibres in a path with two and with three of the sixteen closing edges restored — were re-verified: both layers planar in each case, with join-edge counts $41/39$ and $49/50$, in every case at or below the capacity. A witness exceeding the capacity would refute Lemma 1; none does, and the margin is small, which is why the bound is decisive only once the cycle is closed.

4 Two consequences

A screening test. Lemma 1 is cheap to evaluate and eliminates a candidate before any search: a biplanar graph on N vertices admits no triangle-free subgraph with more than $4N - 8$ edges, and analogous bounds hold for subgraphs of larger girth. The whole family $C_n[K_r]$, $r \geq 4$, is eliminated in one line. A candidate for the Earth–Moon bound must come from a construction whose triangle-free part is small.

On search. Considerable effort was spent searching for a biplanar decomposition of $C_7[K_4]$ with SAT and CEGAR encodings. Such searches find witnesses; they refute nothing unless run to completion, and on control instances known to be non-biplanar they did not terminate. A sharp increase in difficulty was observed when four of the sixteen closing edges were restored to the seven-fibre path, but the count above leaves that configuration feasible — 100 join edges against a capacity of 104 — so the difficulty threshold is not the obstruction. The obstruction is global, and no witness-finding search could have exposed it.

Prior art

The girth-four planar bound is classical, and Beineke's inequalities [3] apply the same reasoning to biplanar graphs directly; Sykora, Szekely and Vrto [8] use a bound of this type at girth ten. Eppstein [5] notes that such methods require the host graph to have no short cycles and gives a different route that tolerates them. The only step here that may not be standard is the application to a subgraph, which is immediate once stated. I have checked [1] and [5] and found no statement of Theorem 1; I have not been able to consult Boutin, Gethner and Sulanke [4] or Gethner's survey [6]. If the result is known, this note should be read as a verification.

References

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